



REPORT REFERENCE:

IMP7903-2

INDUSTRIAL NOISE IMPACT ASSESSMENT

SWIMMING LESSONS & AC UNIT

British Standard 4142: 2014 +A1:2019

CLIENT:

ET Planning

SITE:

30 Allandale Avenue

London

N3 3PJ

SURVEY DATE:

23rd July – 28th July 2025

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1 EXECUTIVE SUMMARY

1.1 Instruction

Impact Acoustics Ltd has been instructed by ET Planning to undertake a background noise survey to British Standard 4142:2014+A1:2019 to determine the impact of break out noise from swimming lessons at Allandale Avenue, London, N3 3PJ, together with the installation of a single air conditioning units located at the rear of the premises at ground floor level.

1.2 Scope of Report

The measurements have been undertaken in accordance with British Standard 4142:2014 +A1:2019 and ISO 1996 – Part 2: 2007 to establish if the proposed site has a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. This report aims to establish the following:

- Existing background noise levels at the Site (L_{A90});
- Noise levels from the Swimming Lessons (L_{Aeq});
- Impact on the nearest noise sensitive property.
- Mitigation Levels if Required

1.3 Summary of Report

1.3.1 Background Noise Level – External on Boundary

Continuous background noise measurements were undertaken between 23rd July and 28th July 2025 at the Assessment Position 1. The daytime background noise levels between 07:00 and 23:00 was found to be $L_{A90,1 \text{ hour}}$ 40 dB. This background noise level has been assessed during daytime periods only as the proposed Swimming Lessons will not operate during night time hours.

Daytime Equivalent Existing Background Noise Levels (07:00 – 23:00)	Measured Levels at Assessment Position 1	$L_{A90, t}$ 40 dB
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1.3.2 Internal Noise Level – Internal During Swimming Lesson at Capacity

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs.

Noise Source	Sound Pressure Level	Highest L_{Amax}	Octave Band Centre Frequency (Hz)						
	dB(A)		63	125	250	500	1k	2k	4k
15:00 – 16:00	66.5	90.2	59.7	54.1	60.4	63.8	62.1	59.5	54.1
16:00 – 17:00	70.7		54.1	52.7	60.9	66.6	67.2	64.1	57.4
17:00 – 18:00	73.2		52.2	49.9	60.1	68.7	69.7	66.9	60.6

Note: The highest of the three $L_{Aeq, 1 \text{ HOUR}}$ dB noise levels have been used as a worst case scenario.

1.4 Noise Sources

1.4.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

Noise Source	Sound Power Level
	dB(A)
Fujitsu AOYG14LMCA	67.0

1.5 BS 4142: 2014 +A1:2019 Assessments

Rating Level	24 dB	9.1	The acoustic feature correction is added to the specific noise level
Background Noise Level $L_{A90,1 \text{ hour}}$	40 dB	8.1	Modal Background Noise Level
Assessment Level	-16 dB	11	The background level is subtracted from the rating level.
Assessment	-16 dB		
Conclusion	The assessment level is 'Low Impact'		

1.6 Swimming Lessons

1.6.1 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB
Background Noise Level	40 dB
Excess Over Background	-2 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

1.6.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	23 dB(A)
BS8233: 2014 criteria	$L_{Aeq \text{ 16 Hour}}$ 30 dB
Excess Over Criteria	-7 dB
Assessment Conclusion	Low Impact

1.7 L_{Amax} dB Assessment

Assessment of L _{Amax} dB Noise Levels	dB(A) Noise Level
Recorded L _{Aeq t} Noise Level	73 dB(A)
Recorded L _{Amax t} Noise Level	90 dB(A)
Calculated Façade Level L _{Aeq t} dB	38 dB(A)
Variance from L _{Aeq t} to L _{Amax t}	17 dB(A)
Calculated Façade Level L _{Amax t} dB	55 dB(A)
Partially Open Window	15 dB(A)
Calculated Internal Noise Level	40 dB(A)
WHO Night Time Criteria	<45 dB(A)
Assessment Conclusion	Achieved

1.8 Car Movements

1.8.1 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB
Background Noise Level	40 dB
Excess Over Background	-6 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

1.8.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	19 dB(A)
BS8233: 2014 criteria	L _{Aeq 16 Hour} 30 dB
Excess Over Criteria	-11 dB
Assessment Conclusion	Low Impact

1.9 Conclusions

The operation of the condenser and the break out noise have been assessed to establish if the swimming lessons will have a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. Measurements have been undertaken in accordance with British Standard 4142: 2014 +A1:2019 and ISO 1996 – Part 2: 2017. This report has established the existing background noise levels at the closest noise sensitive façade to the site and the assessment of the impact of the site operation on nearby residential properties to the east of the site. The resulting emissions from the site running on a worst case scenario show no conflict with ‘Low Impact’ criteria and give a strong indication that complaint and impact on the local amenity is unlikely.

1.10 Conclusion Summary

National Planning Policy Framework 2024 suggests that planning permission should be granted unless any adverse impacts of doing so would significantly and demonstrably outweigh the benefits, when assessed against the policies in the framework taken as a whole, or specific policies in the framework indicate the application should be restricted.

Based on the calculations and assessments made within this report it is the professional opinion of Impact Acoustics Ltd that the proposed development can demonstrate compliance with the National Planning Policy Framework 2024, NPPF & NPSE and that, with regards to sound, planning permission can be granted.

2 INTRODUCTION

2.1 Instruction

Impact Acoustics Ltd has been instructed by ET Planning to undertake a background noise survey to British Standard 4142:2014+A1:2019 to determine the impact of break out noise from swimming lessons at Allandale Avenue, London, N3 3PJ, together with the installation of a single air conditioning units located at the rear of the premises at ground floor level.

The purpose of this assessment is to ensure the acoustic protection of noise sensitive premises closest to the swimming pool. Noise sensitive premises are not restricted to residential dwellings as offices can be affected by unwanted external noise. These have been established as the residential properties to either side of the site.

The swimming lessons will be carried out during daytime hours only and will have a maximum capacity of 4 children and 2 teachers at any one time. These are 30 minute lessons located within the internal pool.

2.2 Ambient and Background Measurements

External noise levels are to be recorded over, what has been considered for the site, an average / typical time period in order to assess the daytime noise levels. Levels have been recorded over more than one day in order to ensure the uncertainty of measurement aspects of BS 4142: 2014 +A1:2019 have been satisfied and that the data recorded is representative for the purpose of a robust assessment.

2.3 BS 4142: 2014 +A1:2019

British Standard 4142: 2014 +A1:2019 is to be adopted for the basis of this background noise level assessment. A BS 4142: 2014 +A1:2019 noise assessment will be carried out in order to demonstrate the acoustic impact the Condensor could have on the nearest affected premises and make suitable recommendations in order to demonstrate this unit will not have a significant and demonstrable adverse impact on the nearest noise sensitive premises in accordance with the National Planning Policy Framework, once remedial works are completed.

2.4 National Planning Policy Framework 2024 & Noise Policy Statement for England 2010

References and evaluations are to be made to the National Planning Policy Framework 2024 (NPPF) and the Noise Policy Statement for England 2010 (NPSE). The purpose of this document is to include all aspects of environmental noise within assessments i.e. environmental noise, neighbour noise and neighbourhood noise. Noise is to be considered alongside other relevant issues relating to the site and should not be considered in isolation, according to the NPSE.

There are several key phrases within the NPSE aims and these are discussed below.

2.5 “Significant adverse” and “adverse”

There are two established concepts from toxicology that are currently being applied to noise impacts, for example, by the World Health Organisation. They are:

2.6 NOEL – No Observed Effect Level

This is the level below which no effect can be detected. In simple terms, below this level, there is no detectable effect on health and quality of life due to the noise.

2.7 LOAEL – Lowest Observed Adverse Effect Level

This is the level above which adverse effects on health and quality of life can be detected.

Extending these concepts for the purpose of this NPSE leads to the concept of a significant observed adverse effect level.

2.8 SOAEL – Significant Observed Adverse Effect Level

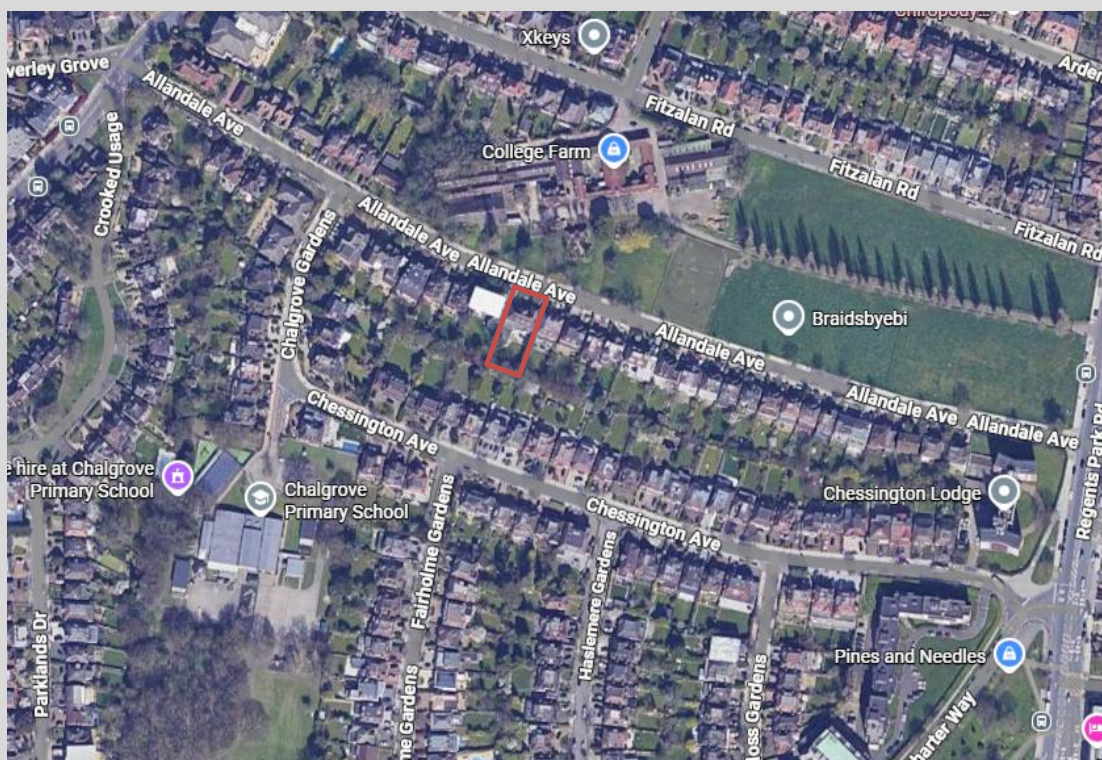
This is the level above which significant adverse effects on health and quality of life occur.

It is not possible to have a single objective noise-based measure that defines SOAEL that is applicable to all sources of noise in all situations. Consequently, the SOAEL is likely to be different for different noise sources, for different receptors and at different times. It is acknowledged that further research is required to increase our understanding of what may constitute a significant adverse impact on health and quality of life from noise. However, not having specific SOAEL values in the NPSE provides the necessary policy flexibility until further evidence and suitable guidance is available.

3 SITE LOCATION

3.1 Location Plan

- Yellow – Background Measurement Position
- Orange NSP (Noise Sensitive Premises)



4 BACKGROUND NOISE LEVELS

4.1 Background Noise Levels

Continuous background noise measurements were undertaken between 23rd July and 28th July 2025 at the Assessment Position 1 externally on the boundary of the property as depicted with the yellow dot in section 3.1 of this report. The daytime background noise levels between 07:00 and 23:00 was found to be $L_{A90,1\text{ hour}}$ 40 dB. This background noise level has been assessed during daytime periods only as the Swimming Lessons will not operate during night time hours. During this period the swimming pool was not in use at all to allow us to capture the prevailing background noise levels for this area.

Daytime Equivalent Existing Background Noise Levels (07:00 – 23:00)	Measured Levels at Assessment Position 1	$L_{A90, t}$ 40 dB
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4.2 Car Movements

This background noise level has also been used to assess the impact of car movement on the front of the site. We recognise the background noise level would be higher at the front of the site, due to the busy road and the noise levels associated with this. However, we have adopted this as a worst case scenario as used this background noise level to assess the potential impact of car movements on the facades at the front of the neighbouring properties.

4.3 Internal Activity

Background noise measurements were made at position 1 externally together with simultaneous measurements inside the swimming pool using a second noise meter, with all doors and windows closed, which is what this assessment has been based on. The swimming pool was used between 15:00 – 18:00 hrs on 24th as detailed in this report and the data extrapolated from the internal noise meter. This was then used as the worst case scenario as the session captured include for maximum capacity of 4 children and two teachers.

5 APPARATUS

5.1 Equipment Calibration

The equipment was calibrated using a sound pressure level of 114.0 dB at an octave band centre frequency of 1000Hz with reference to $2 \times 10^{-5} \text{ Nm}^{-2}$ before and after the tests and the equipment set to have no inaccuracy greater than 0.2 dB.

All the following equipment was calibrated in accordance with the laboratory accreditation requirements of the United Kingdom Accreditation Service (UKAS) on the following dates. Calibration schedules are implemented within Impact Acoustics Ltd in accordance with UKAS directive LAB 23.

5.2 Position 1 – External on Boundary

Description	Make	Type	Serial No.	Calibration Intervals	Cert No	Last Calibrated	Next Due Calibration
Integrated Sound Level Meter	Norsonic	118	31471	2 YEARS	TCRT23-1042	15.01.2024	15.01.2026
12.5mm Microphone (with windshield)	GRAS	40AF	114650	2 YEARS		15.01.2024	15.01.2026
Microphone Pre – Amplifier	Norsonic	1206	30327	2 YEARS		15.01.2024	15.01.2026

5.3 Internal Noise Meter Inside Pool Area

Description	Make	Type	Serial No.	Calibration Intervals	Cert No	Last Calibrated	Next Due Calibration
Integrated Sound Level Meter	Rion	NI-52	242696	2 Years	C00583	26.06.2025	26.06.2027
Microphone (With Windshield)	Rion	Uc-59	6178	2 Years		26.06.2025	26.06.2027
Microphone Pre – Amplifier	Rion	Nh-25	32724	2 Years		26.06.2025	26.06.2027

The noise meters were calibrated before and after the assessment period and found to be within the tolerance of the manufacturer's guidance. Full Calibration certificates are available upon request.

6 PLANT CALCULATIONS

6.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

Noise Source	Sound Power Level
	dB(A)
Fujitsu AOYG14LMCA	67.0

6.2 Noise Breakout Calculations

6.2.1 Roof

Roof	dB(A)	Octave Band Centre Frequency (Hz)						
		63	125	250	500	1	2	4
Internal Noise Level	73.2	52.2	49.9	60.1	68.7	69.7	66.9	60.6
R - Reduction Value		27.6	36.5	38.1	45.5	52.2	51	49.8
S - Surface Area	58.8							
r - Distance (m)	1							
10xLog(s)	17.7	17.7	17.7	17.7	17.7	17.7	17.7	17.7
20xLog r	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
External Calculated Level (Lw2)	46.6	47.3	36.1	44.7	45.9	40.1	38.6	33.5

6.2.2 Rear Wall

Rear Wall	dB(A)	Octave Band Centre Frequency (Hz)						
		63	125	250	500	1	2	4
Internal Noise Level	73.2	52.2	49.9	60.1	68.7	69.7	66.9	60.6
R - Reduction Value		30.6	39.8	39.3	48.7	55.9	55.1	46.8
S - Surface Area	13							
r - Distance (m)	1							
10xLog(s)	11.1	11.1	11.1	11.1	11.1	11.1	11.1	11.1
20xLog r	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
External Calculated Level (Lw2)	37.6	37.8	26.3	36.9	36.1	29.9	28.0	30.0

6.2.3 Side Wall & Windows Combined

Front & Side Walls & Glazing	dB(A)	Octave Band Centre Frequency (Hz)						
		63	125	250	500	1	2	4
Internal Noise Level	73.2	52.2	49.9	60.1	68.7	69.7	66.9	60.6
R - Reduction Value		24.0	32.4	30.3	35.0	39.5	44.7	37.7
S - Surface Area	48							
r - Distance (m)	1							
10xLog(s)	16.8	16.8	16.8	16.8	16.8	16.8	16.8	16.8
20xLog r	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
External Calculated Level (Lw2)	56.3	50.0	39.4	51.6	55.5	52.0	44.0	44.7

6.3 Inputted Noise Levels

Based on the above calculations, the following noise levels have been inputted into CADNA A.

Sound Levels (local)														
OK Cancel Copy... Font... Adjust Col. Width Help														
Name	ID	Type	Oktave Spectrum (dB)											Source
			Weight.	31.5	63	125	250	500	1000	2000	4000	8000	A	lin
RearWall	RearWall	Li		0.0	37.8	26.3	36.9	36.1	29.9	28.0	30.0	0.0	37.6	42.6
Side&FrontWall&Windows	Side&FrontWall&Windows	Li		0.0	50.0	39.4	51.6	55.5	52.0	44.0	44.7	0.0	56.2	59.1
Roof	Roof	Li		0.0	47.3	36.1	44.7	45.9	40.1	38.6	33.5	0.0	46.6	51.6

Inputted Noise Levels

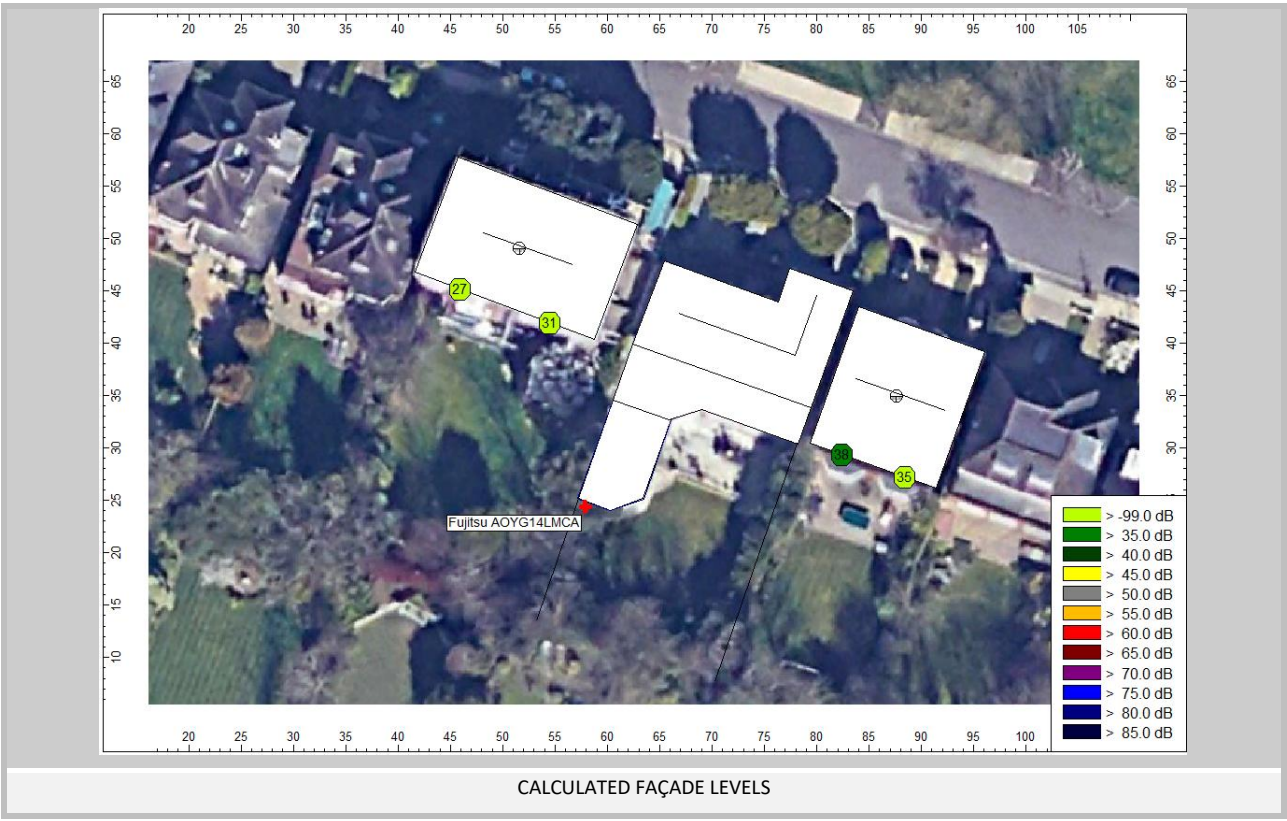
6.4 Calculated Façade Levels

The following demonstrates the above noise levels replicated on the 3D model.

6.4.1 Plant Only (BS4142)



6.4.2 Plant & Swimming Lessons



7 BS4142:2014 +A1:2019 NOISE ASSESSMENT

7.1 Scope of British Standard 4142: 2014

In the assessment of the existing surrounding commercial premises, consideration has been given to the scope of British Standard 4142:2014+A1:2019, which in section 1, details applicability of this standard to rating assessing sound of an industrial and/or commercial nature. It is considered appropriate that both the background noise levels, and the rating noise levels obtained fall within the scope of British Standard 4142:2014+A1:2019 by using outdoor sound levels to assess the effect of sound on local residents.

7.2 Terms and Definitions

Symbol	Term	Definition
AP	Assessment Position	Position externally at the façade property under investigation at which the assessment is undertaken which is usually 1m from the 1 st floor bedroom window.
EP	Equivalent Position	Position at which the background noise levels are measured if there is no access to the assessment position or if source under investigation is audible.
$L_s = L_{Aeq, T}$	Specific Level	The average continuous equivalent sound pressure level of the source at the assessment position.
$L_{Ar, Tr}$	Rating Level	The average continuous equivalent sound pressure level of the source at the assessment position with a correction to account for the characteristic features.
$L_r = L_{Aeq, T}$	Residual Level	The average continuous equivalent sound pressure level at the assessment position without the source operating.
$L_{A90, T}$	Background Level	The sound pressure level that is not exceeded 90% of the time at the assessment position.
$L_a = L_{Aeq, T}$	Ambient Level	The totally encompassing sound at the assessment position including the residual and specific noise.

7.3 Assessment Position

The assessment position 1 was established as the proposed residential accommodation premises to the south of the Pitter Commercial Site.

7.4 Rating Levels (Character Correction)

It is appropriate to add a character correction where there is a new source that cannot be measured in line with BS 4142:2014+A1:2019. There are 3 methods for approaching this.

- Subjective method
- Objective method (for tonality)
- Reference method

7.5 Subjective Method

The subjective method establishes a rating penalty that is added to the specific noise level if any of the following is present at the assessment position. If a tone is expected to be present a character correction of 0 dB to 6 dB is added depending on how perceptible it is at noise sensitive locations.

BS 4142:2014+A1:2019 – Section 9.2 Subjective Method	Perceptibility to noise sensitive facades	Correction
Tonality Ranging from not tonal to prominently tonal	Not tonal	+0
	Just perceptible	+2
	Clearly perceptible	+4
	Highly perceptible	+6

If the source is expected to be impulsive a character correction of 0 dB to 9 dB is added depending on how perceptible it is at noise sensitive locations.

BS 4142:2014+A1:2019 – Section 9.2 Subjective Method	Perceptibility to noise sensitive facades	Correction
Impulsivity Considering both the rapidity and any overall change in sound levels	Not impulsive	+0
	Just perceptible	+3
	Clearly perceptible	+6
	Highly perceptible	+9

When the sound features are neither tonal nor impulsive, a character correction of +3 is added for the readily distinctive quality against the acoustic environment or for the intermittency of the source.

BS 4142:2014+A1:2019 – Section 9.2 Subjective Method	Perceptibility to noise sensitive facades	Correction
Readily Distinctive	Is not present	+0
	Is present	+3
Intermittency	Is not present	+0
	Is present	+3

7.6 Assessment Criterion

The significance of the resulting noise on the residential property depends on the margin by which it exceeds the background noise levels. British Standard 4142:2014+A1:2019 provides the following guidance within section 11.

Difference	Assessment of Impact
+10 dB	Indication of a significant adverse impact
+5 dB	Indication of an adverse impact
+0 dB	Indication of low impact

7.7 Assessments

7.7.1 Existing Daytime (07:00 – 23:00)

Item	Calculation	Clause	Commentary
Specific Noise Level $L_{Aeq,1 \text{ hour}}$	17 dB	7	Calculated using ISO 9613:1996 ^[3] .
Tonal	+2 dB	8.1	Tonality Characteristic
Intermittency	+3 dB	8.1	Intermittency Characteristic
Rating Level	24 dB	9.1	The acoustic feature correction is added to the specific noise level
Background Noise Level $L_{A90,1 \text{ hour}}$	40 dB	8.1	Modal Background Noise Level
Assessment Level	-16 dB	11	The background level is subtracted from the rating level.
Assessment	-16 dB		
Conclusion	The assessment level is 'Low Impact'		

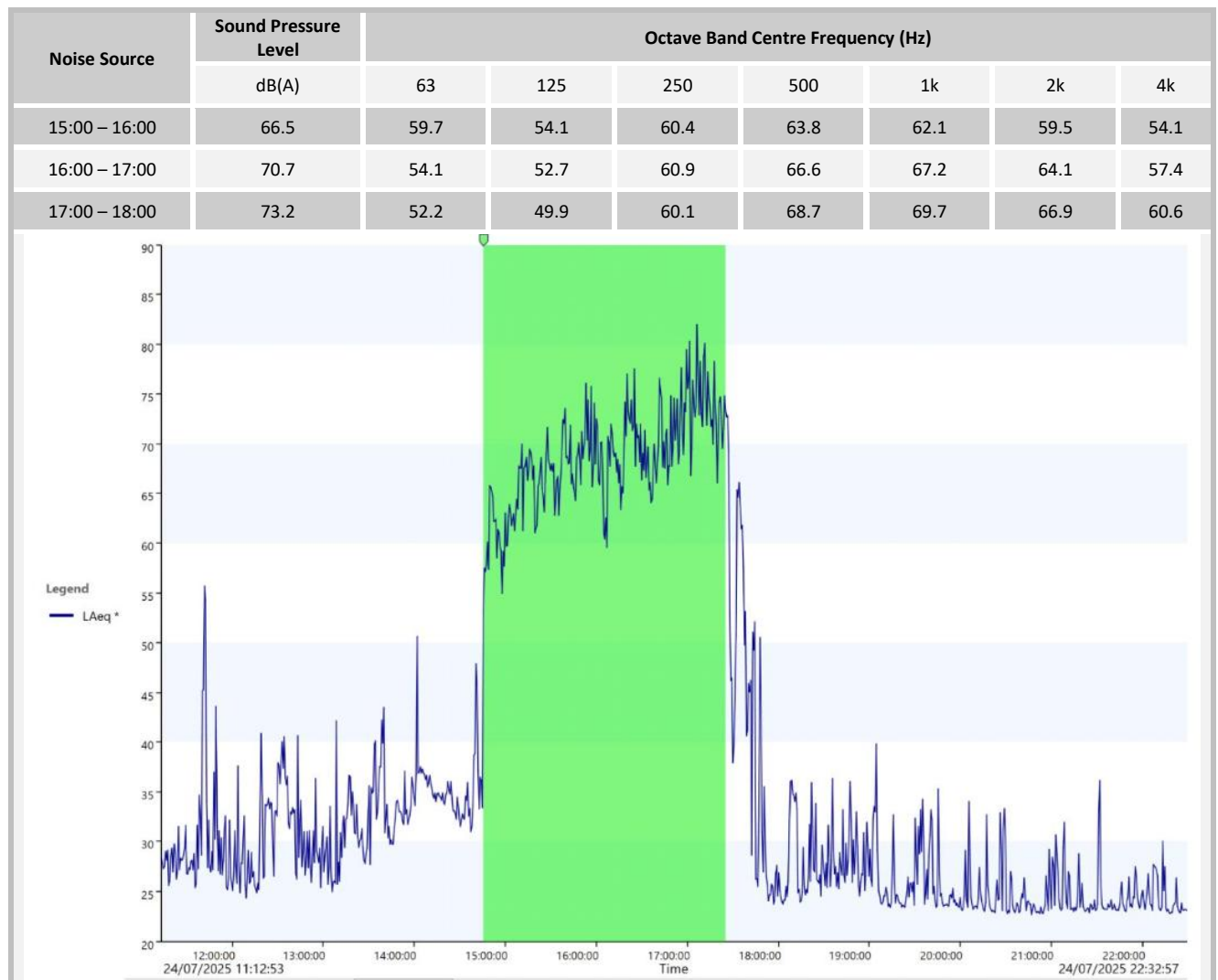
7.8 Assessment Conclusion

It can be seen from the above assessments that an assessment conclusion of Low Impact could be expected. Therefore, no further mitigation measures would be required to the plant at this stage.

8 SWIMMING LESSONS

8.1 Internal Poolside Recorded Noise Levels from Second Internal Noise Meter

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs during a swimming lesson of maximum capacity.



Note: The highest of the three LAeq 1 HOUR dB noise levels have been used as a worst case scenario.

8.2 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB
Background Noise Level	40 dB
Excess Over Background	-2 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

8.3 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

8.3.1 British Standard 8233: 2014

Calculations are to be made in accordance with BS 8233: 2014 Sound Insulation and Noise Reduction for Buildings Code of Practice.

BS 8233: 2014 set the following parameters as target levels that should be designed to within rooms such as Living Rooms and Bedrooms.

Indoor ambient noise levels for dwellings			
Activity	Location	07:00 – 23:00	23:00 – 07:00
Resting	Living Room	35 dB L_{Aeq} 16 HOUR	--
Dining	Dining Room / Area	40 dB L_{Aeq} 16 HOUR	--
Sleeping (daytime resting)	Bedroom	35 dB L_{Aeq} 16 HOUR	30 dB L_{Aeq} 8 HOUR

8.3.2 Calculated Internal Noise level

Calculations and assessments are therefore to be carried out in order to satisfy the above requirements of BS8233: 2014.

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	23 dB(A)
BS8233: 2014 criteria	L_{Aeq} 30 dB
Excess Over Criteria	-7 dB
Assessment Conclusion	Low Impact

9 L_{Amax} dB ASSESSMENT

9.1.1 Outline

In order to satisfy the local council, we have included an L_{Amax} dB assessment within this report in relation to the swimming levels during the lesson monitored between 15:00 – 18:00 hrs a . It should be noted that there is no legislation relating to criteria surrounding L_{Amax} dB levels at facades during the day. The only criteria close to this would be from the World Health Organisation ‘Guidelines for Community Noise’ that suggests in order to not have a detrimental effect on sleep, night time periods from 23:00 – 07:00 hours should have no more than 10 events exceeding L_{Amax} 45 dB. Whilst this can’t be directly attributed to daytime levels, it does offer some level of acoustic value when establishing the likelihood of noise complaints from activities that could be deemed detrimental to neighbouring properties.

9.1.2 Assessment Criterion

The significance of the resulting noise on the residential property depends on the margin by which it exceeds the background noise levels. I consider the following to be appropriate for this assessment.

NPPF 2024 & NPSE 2010	DIFFERENCE	ASSESSMENT OF IMPACT
SOAEL – Significant Observed Adverse Effect Level	+10 dB	Indication of a Significant Adverse Impact
LOAEL – Lowest Observed Adverse Effect Level	+5 dB	Indication of an Adverse Impact
NOEL - No Observed Effect Level	+0 dB	Indication of Low Impact

9.1.3 Calculations

Assessment of L _{Amax} dB Noise Levels	dB(A) Noise Level
Recorded L _{Aeq t} Noise Level	73 dB(A)
Recorded L _{Amax t} Noise Level	90 dB(A)
Calculated Façade Level L _{Aeq t} dB	38 dB(A)
Variance from L _{Aeq t} to L _{Amax t}	17 dB(A)
Calculated Façade Level L _{Amax t} dB	55 dB(A)
Partially Open Window	15 dB(A)
Calculated Internal Noise Level	40 dB(A)
WHO Night Time Criteria	<45 dB(A)
Assessment Conclusion	Achieved

10 CAR MOVEMENTS

10.1 Outline

The swimming lessons are currently being scheduled between 2pm to 8pm in half hour intervals. As the pool only has a capacity of 4 students and 2 teachers, many of the children arrive in the same car. However, there is a theoretical scenario where four cars could all arrive one after the other. Therefore this worst case scenario has been evaluated, although unlikely.

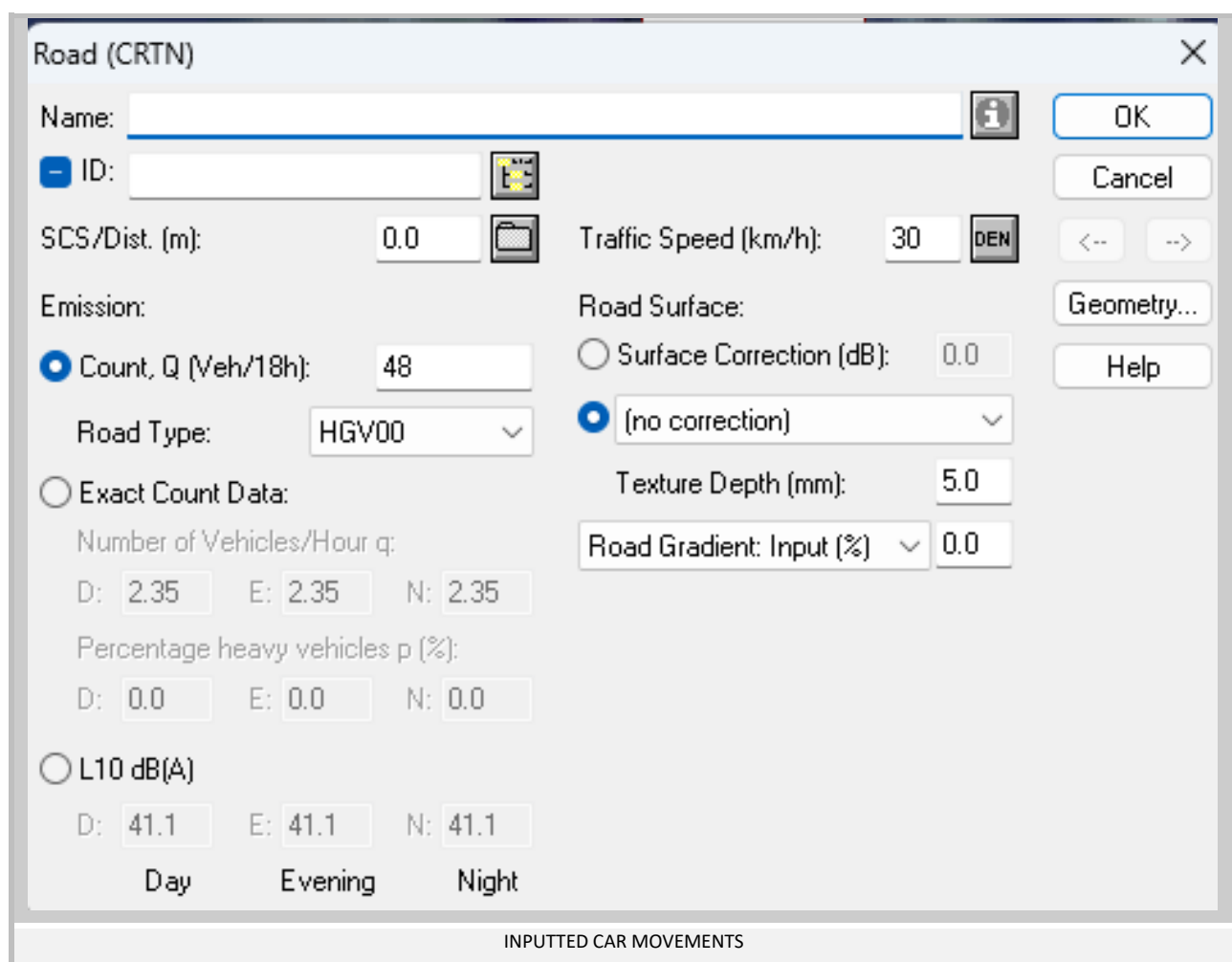
10.2 Calculation

The following calculations demonstrate the maximum flow of cars associated with the swimming pool lessons.

Activity	
Time Period for Lessons	2pm – 8pm (6 Hours)
Number of Cars Per Hour	8
Number of Cars Per Day	48

10.3 Inputted Vehicle Movements

The above 48 car movements have been inputted into the CADNA A Software to calculate the resultant noise levels.



Road (CRTN)

Name:

ID:

SCS/Dist. (m):

Traffic Speed (km/h):

Emission:

☒ Count, Q (Veh/18h):

Road Type:

☐ Exact Count Data:

Number of Vehicles/Hour q:

D: E: N:

Percentage heavy vehicles p (%):

D: E: N:

☐ L10 dB(A)

D: E: N:

Day Evening Night

Road Surface:

☐ Surface Correction (dB):

☒ (no correction)

Texture Depth (mm):

Road Gradient: Input (%)

OK Cancel Geometry... Help

INPUTTED CAR MOVEMENTS

10.4 Calculated Façade Levels

Based on the above, the following has been calculated.



10.5 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB
Background Noise Level	40 dB
Excess Over Background	-6 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of ‘Low Impact’.

10.6 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	19 dB(A)
BS8233: 2014 criteria	$L_{Aeq\ 16\ Hour}$ 30 dB
Excess Over Criteria	-11 dB
Assessment Conclusion	Low Impact

For reference , if the number of swimming lesion and cars were to double to 12 hours and 96 cars, we would expect to see a +3 dB increase in the calculated façade levels. These would still be below that of the prevailing background noise level taken at the rear of the premises.

11 NPPF & NPSE

The National Planning Policy Framework 2024 (NPPF) and assessments to the Noise Policy Statement for England 2010 (NPSE) should be made in conjunction with each other. Paragraphs 198 - 201 of the National Planning Policy Framework 2024 (NPPF) states the following:

Paragraph 198 Planning policies and decisions should also ensure that new development is appropriate for its location taking into account the likely effects (including cumulative effects) of pollution on health, living conditions and the natural environment, as well as the potential sensitivity of the site or the wider area to impacts that could arise from the development. In doing so they should:

- a) Mitigate and reduce to a minimum potential adverse impacts resulting from noise from new development – and avoid noise giving rise to significant adverse impacts on health and the quality of life.
- b) Identify and protect tranquil areas which have remained relatively undisturbed by noise and are prized for their recreational and amenity value for this reason.

Paragraph 199 Planning policies and decisions should sustain and contribute towards compliance with relevant limit values or national objectives for pollutants. So far as possible these opportunities should be considered at the plan-making stage, to ensure a strategic approach and limit the need for issues to be reconsidered when determining individual applications.

Paragraph 200 Planning policies and decisions should ensure that new development can be integrated effectively with existing businesses and community facilities (such as places of worship, pubs, music venues and sports clubs). Existing businesses and facilities should not have unreasonable restrictions placed on them as a result of development permitted after they were established. Where the operation of an existing business or community facility could have a significant adverse effect on new development (including changes of use) in its vicinity, the applicant (or 'agent of change') should be required to provide suitable mitigation before the development has been completed.

Paragraph 201 The focus of planning policies and decisions should be on whether proposed development is an acceptable use of land, rather than the control of processes or emissions (where these are subject to separate pollution control regimes). Planning decisions should assume that these regimes will operate effectively. Equally, where a planning decision has been made on a particular development, the planning issues should not be revisited through the permitting regimes operated by pollution control authorities

The Noise Policy Statement for England gives various levels of effect as detailed within this report.

With the proposed development installed as detailed within this report, the development can be implemented within the guidelines of the aforementioned documents and ensure a conclusion of **NOEL – No Observed Effect Level**. This is the level below which no effect can be detected. In simple terms, below this level, there is no detectable effect on health and quality of life due to the noise.

12 CONCLUSIONS

12.1 Summary of Report

12.1.1 Background Noise Level – External on Boundary

Continuous background noise measurements were undertaken between 23rd July and 28th July 2025 at the Assessment Position 1. The daytime background noise levels between 07:00 and 23:00 was found to be $L_{A90,1 \text{ hour}}$ 40 dB. This background noise level has been assessed during daytime periods only as the proposed Swimming Lessons will not operate during night time hours.

Daytime Equivalent Existing Background Noise Levels (07:00 – 23:00)	Measured Levels at Assessment Position 1	$L_{A90, t}$ 40 dB
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12.1.2 Internal Noise Level – Internal During Swimming Lesson at Capacity

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs.

Noise Source	Sound Pressure Level	Highest L_{Amax}	Octave Band Centre Frequency (Hz)						
	dB(A)	dB(A)	63	125	250	500	1k	2k	4k
15:00 – 16:00	66.5	90.2	59.7	54.1	60.4	63.8	62.1	59.5	54.1
16:00 – 17:00	70.7		54.1	52.7	60.9	66.6	67.2	64.1	57.4
17:00 – 18:00	73.2		52.2	49.9	60.1	68.7	69.7	66.9	60.6

Note: The highest of the three $L_{Aeq 1 \text{ HOUR}}$ dB noise levels have been used as a worst case scenario.

12.2 Noise Sources

12.2.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

Noise Source	Sound Power Level
	dB(A)
Fujitsu AOYG14LMCA	67.0

12.3 BS 4142: 2014 +A1:2019 Assessments

Rating Level	24 dB	9.1	The acoustic feature correction is added to the specific noise level
Background Noise Level $L_{A90,1 \text{ hour}}$	40 dB	8.1	Modal Background Noise Level
Assessment Level	-16 dB	11	The background level is subtracted from the rating level.
Assessment	-16 dB		
Conclusion	The assessment level is 'Low Impact'		

12.4 Swimming Lessons

12.4.1 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB
Background Noise Level	40 dB
Excess Over Background	-2 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

12.4.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	38 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	23 dB(A)
BS8233: 2014 criteria	$L_{Aeq \text{ 16 Hour}}$ 30 dB
Excess Over Criteria	-7 dB
Assessment Conclusion	Low Impact

12.5 L_{Amax} dB Assessment

Assessment of L_{Amax} dB Noise Levels	dB(A) Noise Level
Recorded $L_{Aeq \text{ } t}$ Noise Level	73 dB(A)
Recorded $L_{Amax \text{ } t}$ Noise Level	90 dB(A)
Calculated Façade Level $L_{Aeq \text{ } t}$ dB	38 dB(A)
Variance from $L_{Aeq \text{ } t}$ to $L_{Amax \text{ } t}$	17 dB(A)
Calculated Façade Level $L_{Amax \text{ } t}$ dB	55 dB(A)
Partially Open Window	15 dB(A)
Calculated Internal Noise Level	40 dB(A)
WHO Night Time Criteria	<45 dB(A)
Assessment Conclusion	Achieved

12.6 Car Movements

12.6.1 Assessments

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB
Background Noise Level	40 dB
Excess Over Background	-6 dB
Assessment Conclusion	Low Impact

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

12.6.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

External Residential Assessment	Calculated Façade Level
Calculated Level at Residential	34 dB(A)
Partially Open Window	15 dB(A)
Calculated internal Noise level	19 dB(A)
BS8233: 2014 criteria	L _{Aeq} 16 Hour 30 dB
Excess Over Criteria	-11 dB
Assessment Conclusion	Low Impact

12.7 Conclusions

The operation of the condenser and the break out noise have been assessed to establish if the swimming lessons will have a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. Measurements have been undertaken in accordance with British Standard 4142: 2014 +A1:2019 and ISO 1996 – Part 2: 2017. This report has established the existing background noise levels at the closest noise sensitive façade to the site and the assessment of the impact of the site operation on nearby residential properties to the east of the site. The resulting emissions from the site running on a worst case scenario show no conflict with 'Low Impact' criteria and give a strong indication that complaint and impact on the local amenity is unlikely.

12.8 Conclusion Summary

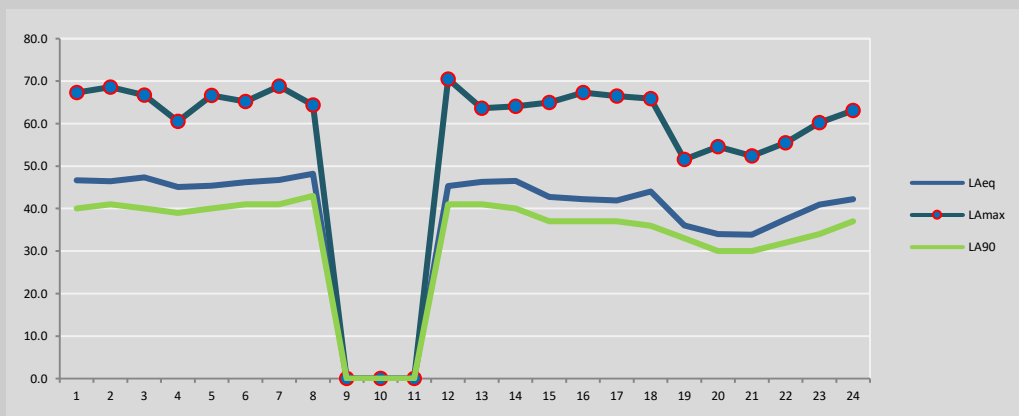
National Planning Policy Framework 2024 suggests that planning permission should be granted unless any adverse impacts of doing so would significantly and demonstrably outweigh the benefits, when assessed against the policies in the framework taken as a whole, or specific policies in the framework indicate the application should be restricted.

Based on the calculations and assessments made within this report it is the professional opinion of Impact Acoustics Ltd that the proposed development can demonstrate compliance with the National Planning Policy Framework 2024, NPPF & NPSE and that, with regards to sound, planning permission can be granted.

13 APPENDIX A – RESULTS

13.1 23rd July – 24th July 2025

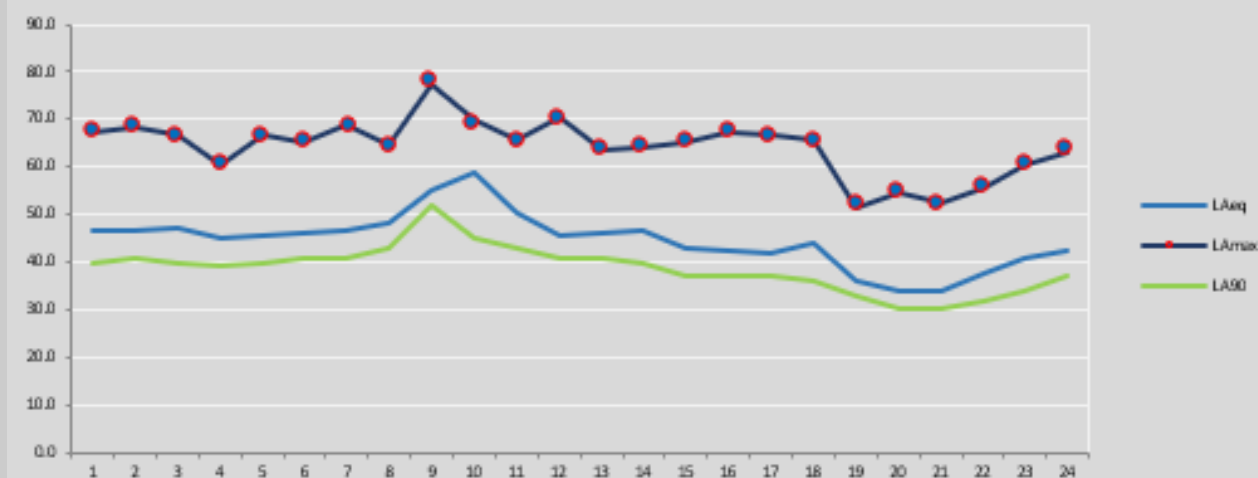
Time	L _{Max}	Minimum 15 minute		1 Hour		Day and Night	
Period		L _{Aeq,15 min}	L _{A90, 15 min}	L _{Aeq, 1 hour}	L _{A90, 1 hour}	L _{Aeq, t}	L _{A90, 16hours}
07:00 – 08:00	0.0	0.0	0.0	0.0	0.0	48.6	38.0
08:00 – 09:00	0.0	0.0	0.0	0.0	0.0		
09:00 – 10:00	0.0	0.0	0.0	0.0	0.0		
10:00 – 11:00	0.0	0.0	0.0	0.0	0.0		
11:00 – 12:00	0.0	0.0	0.0	0.0	0.0		
12:00 – 13:00	75.9	47.1	0.0	46.9	36.0		
13:00 – 14:00	75.4	45.3	38.0	56.1	40.0		
14:00 – 15:00	71.5	49.6	39.0	56.3	41.0		
15:00 – 16:00	68.7	46.4	40.0	47.5	40.0		
16:00 – 17:00	65.5	45.8	40.0	46.4	41.0		
17:00 – 18:00	64.8	43.6	38.0	44.9	40.0		
18:00 – 19:00	61.0	44.5	39.0	45.6	40.0		
19:00 – 20:00	73.5	43.8	39.0	47.3	40.0		
20:00 – 21:00	62.0	41.0	38.0	43.1	39.0		
21:00 – 22:00	62.0	43.1	38.0	43.9	39.0		
22:00 – 23:00	58.5	41.9	39.0	42.9	40.0	42.7	37.0
23:00 – 00:00	55.9	40.4	38.0	41.8	39.0		
00:00 – 01:00	58.7	40.5	38.0	41.2	39.0		
01:00 – 02:00	58.8	37.8	36.0	40.7	37.0		
02:00 – 03:00	63.1	35.8	34.0	40.4	35.0		
03:00 – 04:00	51.2	35.3	33.0	35.5	34.0		
04:00 – 05:00	60.0	34.2	33.0	38.2	33.0		
05:00 – 06:00	63.3	40.1	36.0	45.1	38.0		
06:00 – 07:00	65.4	43.1	38.0	42.9	40.0		



Sound Pressure Levels, dB. Reference level of $2 \times 10^{-5} \text{ Nm}^{-2}$

13.2 24th July – 25th July 2025

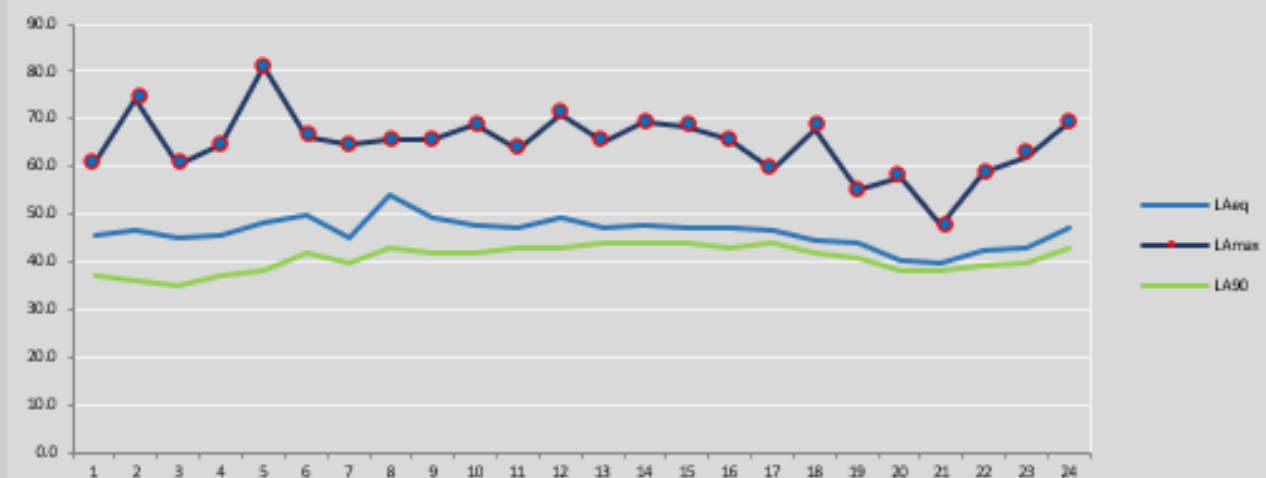
Time	L _{Max}	Minimum 15 minute		1 Hour		Day and Night	
Period		L _{Aeq,15 min}	L _{A90, 15 min}	L _{Aeq, 1 hour}	L _{A90, 1 hour}	L _{Aeq, t}	L _{A90, 16hours}
07:00 – 08:00	67.3	44.5	40.0	46.7	40.0	45.1	40.0
08:00 – 09:00	68.6	45.5	40.0	46.5	41.0		
09:00 – 10:00	66.7	43.8	39.0	47.3	40.0		
10:00 – 11:00	60.5	43.7	39.0	45.1	39.0		
11:00 – 12:00	66.6	43.4	40.0	45.4	40.0		
12:00 – 13:00	65.2	44.4	41.0	46.2	41.0		
13:00 – 14:00	68.8	45.7	41.0	46.7	41.0		
14:00 – 15:00	64.4	47.0	42.0	48.2	43.0		
15:00 – 16:00	0.0	0.0	0.0	0.0	0.0		
16:00 – 17:00	0.0	0.0	0.0	0.0	0.0		
17:00 – 18:00	0.0	0.0	0.0	0.0	0.0		
18:00 – 19:00	70.5	44.2	41.0	45.3	41.0		
19:00 – 20:00	63.6	43.6	40.0	46.3	41.0		
20:00 – 21:00	64.1	42.7	38.0	46.5	40.0		
21:00 – 22:00	65.0	38.7	37.0	42.7	37.0	42.0	35.0
22:00 – 23:00	67.3	38.5	37.0	42.2	37.0		
23:00 – 00:00	66.5	38.7	36.0	41.9	37.0		
00:00 – 01:00	65.9	38.1	35.0	44.0	36.0		
01:00 – 02:00	51.6	33.8	33.0	36.0	33.0		
02:00 – 03:00	54.6	31.0	30.0	34.0	30.0		
03:00 – 04:00	52.4	32.1	29.0	33.9	30.0		
04:00 – 05:00	55.5	32.0	31.0	37.5	32.0		
05:00 – 06:00	60.2	35.0	32.0	41.0	34.0		
06:00 – 07:00	63.1	43.9	36.0	42.2	37.0		



Sound Pressure Levels, dB. Reference level of $2 \times 10^{-5} \text{ Nm}^{-2}$

13.3 25th July – 26th July 2025

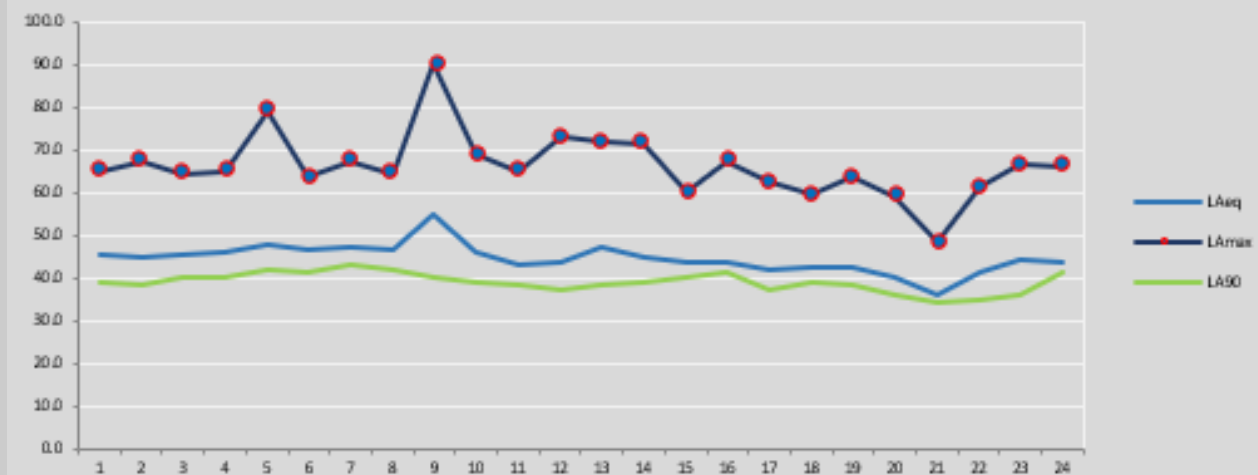
Time	L _{Max}	Minimum 15 minute		1 Hour		Day and Night	
Period		L _{Aeq,15 min}	L _{A90, 15 min}	L _{Aeq, 1 hour}	L _{A90, 1 hour}	L _{Aeq, t}	L _{A90, 16hours}
07:00 – 08:00	60.9	43.7	36.0	45.8	37.0	48.4	42.0
08:00 – 09:00	74.0	43.6	35.0	46.8	36.0		
09:00 – 10:00	60.6	42.2	35.0	44.9	35.0		
10:00 – 11:00	64.8	44.3	37.0	45.8	37.0		
11:00 – 12:00	81.1	46.5	37.0	48.1	38.0		
12:00 – 13:00	66.3	43.0	38.0	49.6	42.0		
13:00 – 14:00	64.8	43.0	40.0	44.8	40.0		
14:00 – 15:00	65.5	48.2	41.0	54.1	43.0		
15:00 – 16:00	65.8	44.4	41.0	49.3	42.0		
16:00 – 17:00	68.7	45.8	41.0	47.9	42.0		
17:00 – 18:00	63.7	46.3	42.0	47.1	43.0		
18:00 – 19:00	71.2	47.5	43.0	49.4	43.0		
19:00 – 20:00	65.0	46.6	44.0	47.2	44.0		
20:00 – 21:00	69.5	46.8	43.0	47.4	44.0		
21:00 – 22:00	68.4	46.4	44.0	47.4	44.0		
22:00 – 23:00	65.7	45.6	43.0	47.0	43.0	43.9	41.0
23:00 – 00:00	59.5	45.3	43.0	46.4	44.0		
00:00 – 01:00	67.9	43.9	41.0	44.6	42.0		
01:00 – 02:00	54.9	42.7	40.0	43.8	41.0		
02:00 – 03:00	57.8	38.8	37.0	40.4	38.0		
03:00 – 04:00	47.6	39.5	38.0	39.8	38.0		
04:00 – 05:00	59.0	40.3	38.0	42.2	39.0		
05:00 – 06:00	62.2	42.2	39.0	43.0	40.0		
06:00 – 07:00	69.5	46.0	40.0	47.0	43.0		



Sound Pressure Levels, dB. Reference level of $2 \times 10^{-5} \text{ Nm}^{-2}$

13.4 26th July – 27th July 2025

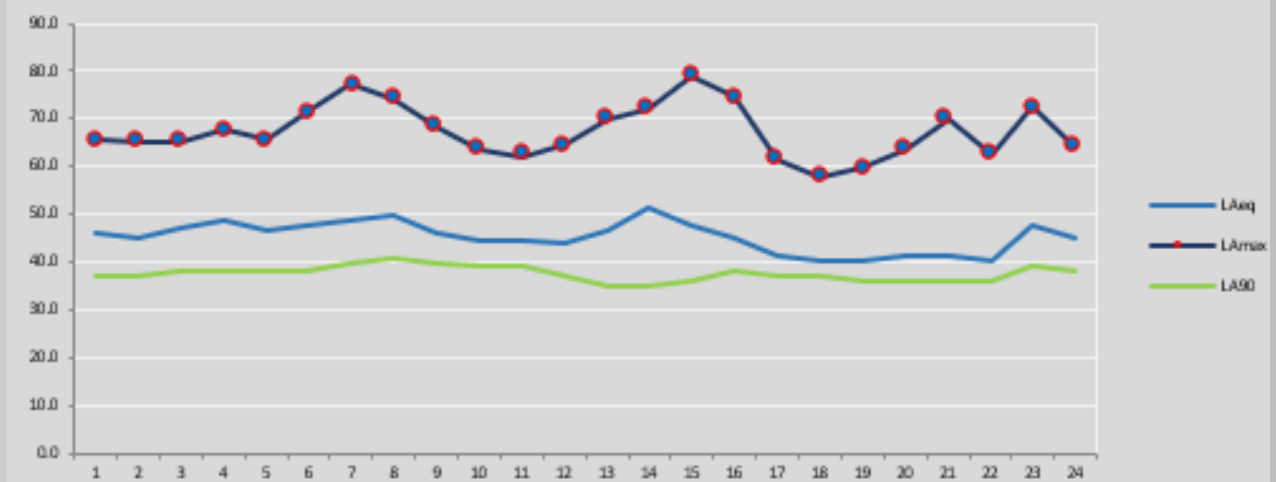
Time	L _{Max}	Minimum 15 minute		1 Hour		Day and Night	
Period		L _{Aeq,15 min}	L _{A90, 15 min}	L _{Aeq, 1 hour}	L _{A90, 1 hour}	L _{Aeq, t}	L _{A90, 16hours}
07:00 – 08:00	65.0	43.7	38.0	45.1	39.0	47.1	40.0
08:00 – 09:00	67.2	42.3	38.0	44.6	38.0		
09:00 – 10:00	64.2	43.2	39.0	45.1	40.0		
10:00 – 11:00	64.9	43.6	39.0	45.7	40.0		
11:00 – 12:00	79.1	44.8	42.0	47.7	42.0		
12:00 – 13:00	63.4	44.5	40.0	46.2	41.0		
13:00 – 14:00	67.0	45.8	42.0	47.3	43.0		
14:00 – 15:00	64.2	45.5	41.0	46.8	42.0		
15:00 – 16:00	90.2	47.1	39.0	54.7	40.0		
16:00 – 17:00	68.7	41.7	39.0	45.9	39.0		
17:00 – 18:00	64.8	40.6	37.0	43.1	38.0		
18:00 – 19:00	73.1	40.0	37.0	43.4	37.0		
19:00 – 20:00	72.0	42.4	37.0	46.9	38.0		
20:00 – 21:00	71.5	42.1	38.0	44.5	39.0		
21:00 – 22:00	60.3	41.6	39.0	43.4	40.0	42.4	37.0
22:00 – 23:00	67.4	42.3	40.0	43.4	41.0		
23:00 – 00:00	62.3	39.4	37.0	42.0	37.0		
00:00 – 01:00	59.3	41.5	38.0	42.3	39.0		
01:00 – 02:00	63.6	39.8	37.0	42.5	38.0		
02:00 – 03:00	58.6	36.3	35.0	39.8	36.0		
03:00 – 04:00	48.4	35.2	34.0	36.0	34.0		
04:00 – 05:00	61.2	36.7	35.0	40.9	35.0		
05:00 – 06:00	66.3	37.2	35.0	44.0	36.0		
06:00 – 07:00	65.9	44.8	36.0	43.4	41.0		



Sound Pressure Levels, dB. Reference level of $2 \times 10^{-5} \text{ Nm}^{-2}$

13.5 27th July – 28th July 2025

Time	L _{Max}	Minimum 15 minute		1 Hour		Day and Night	
Period		L _{Aeq,15 min}	L _{A90, 15 min}	L _{Aeq, 1 hour}	L _{A90, 1 hour}	L _{Aeq, t}	L _{A90, 16hours}
07:00 – 08:00	65.7	43.1	36.0	45.8	37.0	47.4	38.0
08:00 – 09:00	65.2	40.6	35.0	45.0	37.0		
09:00 – 10:00	65.0	46.5	37.0	47.2	38.0		
10:00 – 11:00	67.7	47.3	37.0	48.8	38.0		
11:00 – 12:00	65.8	45.3	37.0	46.8	38.0		
12:00 – 13:00	71.6	44.7	38.0	47.5	38.0		
13:00 – 14:00	77.3	46.6	39.0	49.0	40.0		
14:00 – 15:00	74.3	48.0	40.0	49.7	41.0		
15:00 – 16:00	68.3	44.3	39.0	46.0	40.0		
16:00 – 17:00	63.8	44.1	39.0	44.6	39.0		
17:00 – 18:00	62.2	43.6	38.0	44.4	39.0		
18:00 – 19:00	64.5	41.3	36.0	44.2	37.0		
19:00 – 20:00	69.9	39.7	34.0	46.4	35.0		
20:00 – 21:00	72.0	47.3	35.0	51.5	35.0		
21:00 – 22:00	79.2	40.6	35.0	47.7	36.0		
22:00 – 23:00	74.6	39.6	36.0	45.0	38.0	43.5	38.0
23:00 – 00:00	61.7	40.0	37.0	41.5	37.0		
00:00 – 01:00	57.5	39.5	37.0	40.1	37.0		
01:00 – 02:00	59.8	37.9	34.0	40.2	36.0		
02:00 – 03:00	63.4	38.3	36.0	41.2	36.0		
03:00 – 04:00	70.1	38.6	36.0	41.4	36.0		
04:00 – 05:00	62.6	37.2	35.0	40.1	36.0		
05:00 – 06:00	72.4	41.5	38.0	47.9	39.0		
06:00 – 07:00	64.2	46.1	42.0	45.0	38.0		

Sound Pressure Levels, dB. Reference level of $2 \times 10^{-5} \text{ Nm}^{-2}$

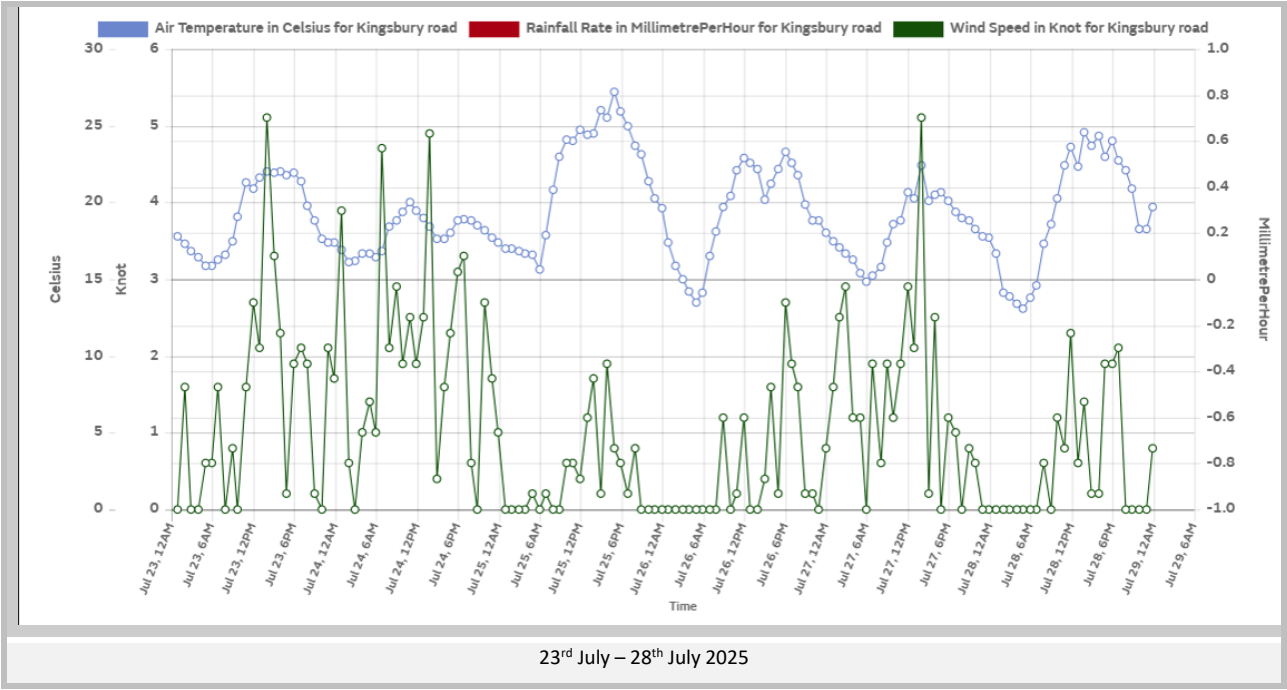
14 APPENDIX B - UNCERTAINTY

In line with the requirements of section 10 of British Standard 4142: 2014 it is expected that the reported expanded uncertainty with a confidence limit of 95% and assuming a convergence of $k=2$ is likely to be ± 2.1 dB. Uncertainty, in this instance has been minimised by undertaking longer background noise measurements over a 120-hour period.

14.1 Uncertainty Budget

Sources of Uncertainty	Uncertainty Notes	Commentary	Value (half Width)	Convert to Same Units (dB)	Distribution Divisor			Standard Uncertainty (u) dB
					Normal	Rectangular	Other	
Measurement Position	Choice of position, ,	1m in 20m		0.2		rect($\sqrt{3}$)		0.13
	microphone orientation	Type 1 0 - 30deg	0.4		Normal			0.50
Instrumentation	Calibration	Calibration Drift	0.1		Normal			0.10
	Accuracy and precision (type 1)	Type 1 practical	1.9			rect($\sqrt{3}$)		0.50
Background Noise Level	Timing of Measurement							
Background Noise Level	Modal Analysis Day	Calculated Standard Deviation	3.7				s/ $\sqrt{n}$	0.39
Background Noise Level	Modal Analysis Night	Calculated Standard Deviation	2.2				s/ $\sqrt{n}$	0.16
Combined Uncertainty (root sum of squares)								1.05
Expanded Uncertainty $U = Kuc$ (95% Confidence $K=2$)								2.1
Final Answer Expressed as Value $\pm U$ dB with a confidence Level of 95%								
REFERENCE: Uncertainty Budget Calculated in line with M3003: The Expression of Uncertainty and Confidence in Measurement Edition 3, November 2012 and A Good Practice Guide on the Sources and Magnitude of Uncertainty Arising in the Practical Measurement of Environmental Noise N J Craven, G Kerry Edition 1a – May 2007.								

15 APPENDIX C – WEATHER CONDITIONS



16 APPENDIX D – BS 4142:2019+A1:2019 INFORMATION

In accordance with BS 4142:2014:A1:2019, section 12, the following information is provided.

16.1 Professional Competence

Acoustic Consultant	Mr Brian Scrivener MIOA
Qualifications	Member of the Institute of Acoustics (MIOA)
Date Certified	June 2003
Time Operating as an active Noise Consultant	Permanently Since June 2003
Professional Status	Noise Consultant
Professional Statement	
I have been active within the field of noise consultancy and noise control engineering since I joined the company in April 1997. Working my way through the company and gaining experience in this specialist field, I qualified as a noise consultant in June 2003 when I passed my Diploma in Noise Control & Engineering from Epsom College under the tutor of Dr Latha Vesudevan. Subsequently I was awarded the status of 'Member of the Institute of Acoustics (MIOA)'. Since then, I have taken over the company and continued to work and develop my professional competence within the field of acoustic consultancy, engineering and noise control.	

16.2 Site Status

Site Attended By	Mr Brian Scrivener MIOA
Baseline Measurements Conducted by	Mr Brian Scrivener MIOA
Site Source Measurements Conducted By	Mr Brian Scrivener MIOA

16.3 Measurement Information – Location 1

Details & Justification	Measurement position 1 was located at the rear of the existing premises, representative of the Nearest Noise Sensitive Premises
Height from Ground	1.5m
Reflective Surfaces	None
Distance to NSP	15m
Distance to Source	10m
Topographical Change (m)	None
Ground between Source and Receiver	100% Soft
Windshield Information	Norsonic Outdoor Microphone Protection Kit Model No: 1212
Distance from Source to Receiver	15m
NSP* - Noise Sensitive Premises	

END OF REPORT